



KrishiDisha

কৃষিদিশা

Agricultural Intelligence & Decision Support Platform

Domain	Agriculture, Education, Nutrition, E-Commerce, Tourism
Target Users	Admin, Farmer, Dealer, Tourist, Cook, Expert, Guide, General User
Tech Stack	PHP, MySQL, HTML, CSS, Bootstrap, XAMPP
Course	Database Management Systems (DBMS)

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1. Project Overview

KrishiDisha (কৃষিদিশা) is a comprehensive, database-driven agricultural intelligence platform that unifies crop knowledge, marketplace trading, tourism management, nutrition science, expert consultation, and administrative oversight into a single web-based ecosystem. The system is designed to serve Bangladesh's agricultural sector by digitising farm-to-market flows, empowering farmers with decision support tools, attracting agri-tourists, and connecting nutritional food culture with modern users.

Vision Statement

To build a one-stop agricultural intelligence platform where every stakeholder — from a subsistence farmer in Rajshahi to an agri-tourist visiting from abroad — can access, trade, learn, and grow.

1.1 Core Modules

#	Module	Core Feature	Primary Beneficiaries
1	Crop Encyclopedia	Full crop intelligence with images, scientific data, varieties, history	Students, Researchers, Farmers
2	Marketplace	Farmer-to-Dealer-to-User product trading with admin commission	Farmers, Dealers, General Users
3	Disease Detection	Image-based disease ID with symptoms and solutions	Farmers, Agricultural Officers
4	Crop Recommendation	Region/soil/season-based smart suggestions	Beginner Farmers
5	Nutrition & Vitamins	Vitamin-based food and crop guidance	Health-conscious Users, Tourists
6	Cooking & Authentic Food	Recipe system with nutrient retention; Cook service for tourists	Tourists, Cooks, General Users
7	Tourism & Farm Visits	Land visits, guide hiring, farm experience booking	Tourists, Farmers, Guides
8	Expert Consultation	Paid consultation between farmers and agricultural experts	Farmers, Experts
9	Profit Calculator	Investment-to-profit analysis for crops	Farmers, Agri-researchers
10	Admin Dashboard	Commission tracking, system management, analytics	Admin

2. System Roles & Detailed Responsibilities

KrishiDisha operates on a role-based access control (RBAC) model. Eight distinct roles exist within the system, each with specific permissions, capabilities, and interaction pathways. All roles extend from a unified USER table.

2.1 Admin

Role Summary

The Admin is the central authority of the KrishiDisha platform. The Admin does not participate in buying, selling, or farming activities directly, but governs all content, approves entities, tracks commissions, and ensures system integrity.

2.1.1 Authentication & Access

- Secure login with hashed password and admin-level privileges
- Role: admin — highest privilege tier
- Access to all modules: read, write, delete, approve, reject

2.1.2 Content Management

- Add, edit, delete crop encyclopedia entries including scientific names, origin, history, images
- Manage crop varieties, diseases (with images, symptoms, solutions), vitamins, and recipes
- Update country-wise crop demand and trade status data
- Manage region-soil-season recommendation mappings

2.1.3 User & Entity Management

- Approve or reject Dealer registrations and verify Farmer accounts
- Manage Expert profiles and set consultation visibility
- Activate or deactivate Cook and Guide accounts
- Monitor tourist accounts and booking activity

2.1.4 Commission & Revenue Management

Transaction Type	Commission Source	Admin Action
Marketplace Sale	% of dealer's total sale amount	Set commission rate per dealer category
Farm Tour Booking	% of total booking fee (land + guide)	Set tourism commission rate
Expert Consultation	% of consultation fee paid by farmer	Set consultation commission rate
Authentic Food Order	% of cook's food service fee	Set food service commission rate

2.1.5 Analytics & Reporting

- View total revenue, commission earned, and pending transactions
- Monitor crop popularity, disease reports, and regional farming trends
- Generate reports on marketplace volume, tourist visits, and consultation activity

2.2 Farmer

Role Summary

Farmers are the primary producers in the KrishiDisha ecosystem. They list products, manage farm lands, hire consultants, and open their farms for agri-tourists. Farmers benefit most directly from the crop recommendation and profit calculator modules.

2.2.1 Account & Profile

- Register with farm name, district, land area, soil type
- Profile verified by Admin before full marketplace access is granted
- Can update farm details, soil type, and growing season information

2.2.2 Marketplace — Product Listing

- List crops/produce with quantity (kg), price per kg, product description, and image
- Manage listing status: active, sold, expired
- View incoming orders from Dealers and track shipment status
- Receive payment via bKash, Nagad, Rocket, card, or bank transfer

2.2.3 Farm Land & Tourism

- Register farm land plots with name, location, area, description, and photo
- Set per-visit fee and mark land as open or closed for tourist visits
- View upcoming tour bookings on their land
- Earn revenue from tourist visit fees (admin takes a fixed %)

2.2.4 Expert Consultation

- Browse available Experts filtered by specialisation and rating
- Book a consultation session with date, time, and duration
- Describe farming problem in notes; receive expert reply after session
- Pay consultation fee through supported payment methods

2.2.5 Crop Tools

- Use Crop Recommendation System: input district/soil/season → get suitable crop suggestions
- Use Profit Calculator: input land area, investment, expected yield, market price → get profit estimate
- View disease alerts and solutions for crops they grow

2.3 Dealer

Role Summary

Dealers act as middlemen in the KrishiDisha marketplace. They purchase produce from Farmers and sell to general Users, earning a margin. Dealers pay a commission to the Admin on every completed sale. They must be approved by the Admin before they can trade.

2.3.1 Registration & Approval

- Register with company name, trade license number, and contact details
- Account is inactive until Admin approves the license and verifies the entity
- Commission rate is assigned by Admin at the time of approval

2.3.2 Buying from Farmers

- Browse the Marketplace and filter by crop type, region, price, and availability
- Place purchase orders on Farmer listings with desired quantity
- Track order status: pending → confirmed → shipped → delivered
- Build inventory from purchased farmer stock (DEALER_INVENTORY table)

2.3.3 Selling to Users

- List products from their inventory to general users at a marked-up sell price
- Manage stock levels: stock quantity auto-decrements on each confirmed sale
- Set sell price, unit, and product description for their listings

2.3.4 Commission & Payments

- On each completed sale: admin_commission (%) is auto-calculated from total_amount
- Dealer net revenue = total_amount – admin_commission
- All payments recorded in PAYMENT table with transaction ID and method
- Commission ledger visible in dealer dashboard for transparency

2.4 Tourist

Role Summary

Tourists are visitors — local or international — who wish to experience Bangladesh's agricultural landscape. They book farm tours, hire guides, order authentic local food prepared by Cooks, and explore the nutritional heritage of Bangladeshi produce.

2.4.1 Registration & Profile

- Register with nationality, passport number, and visit date range
- Can browse the platform freely; booking requires login

2.4.2 Farm Tour Booking

- Browse available FARM_LAND listings with photos, location, area, and visit fee
- Filter by district, crop type grown, or season
- Book a tour: select land, visit date, number of visitors, and optionally hire a Guide
- Total fee = land_fee + guide_fee; admin commission deducted automatically
- Booking status: pending → confirmed → visited → cancelled

2.4.3 Guide Hiring

- Browse available Guides filtered by language, rating, and daily rate
- Guides are assigned to a tour booking; not hired independently
- Guide fee paid as part of the tour booking payment

2.4.4 Authentic Food Experience

- Browse Recipes linked to authentic Bangladeshi crops
- Select a recipe and a Cook who offers that dish
- Place a TOURIST_FOOD_ORDER with special requests if needed
- Pay the Cook's session price; Cook prepares and serves the food
- Admin receives a % commission from the Cook's food service fee

2.4.5 Nutritional Exploration

- Explore Vitamin-based food recommendations: choose a vitamin → see crops → see recipes
- View nutrient retention data for different cooking methods
- Access crop encyclopedia for educational value during visits

2.5 Cook

Role Summary

Cooks are culinary professionals or skilled home cooks registered on KrishiDisha to provide authentic food experiences for tourists. They create and manage recipes, set their availability, and fulfil food orders placed by tourists.

2.5.1 Registration & Profile

- Register as a Cook with speciality (e.g., Chittagong seafood, Sylheti cuisine), location, and price per session
- Mark availability for tourist bookings
- Build a rating through completed food orders

2.5.2 Recipe Management

- Create recipes linked to specific crops in the encyclopedia
- Specify: recipe name, ingredients, step-by-step cooking instructions, cooking method (boil/steam/raw/fry/bake)
- Record nutrient retention percentage — guides tourists on healthiest preparation
- Mark recipes as 'authentic' to highlight traditional dishes

- Upload recipe images

2.5.3 Order Fulfilment

- Receive TOURIST_FOOD_ORDER notifications in dashboard
- Confirm, prepare, and mark order as served
- View special dietary requests submitted by tourists
- Payment received automatically on order confirmation; admin commission deducted

2.6 Agricultural Expert / Consultant

Role Summary

Experts are certified agricultural consultants, agronomists, or researchers who provide paid advisory services to farmers. They receive consultation requests, review farmer notes, and provide written or live expert replies.

2.6.1 Profile & Availability

- Register with specialisation (soil science, pest control, crop disease, irrigation, etc.)
- Set hourly rate and mark availability status
- Build public rating through completed consultations

2.6.2 Consultation Workflow

1. Farmer submits a consultation request with date, duration, and problem description
2. Expert receives notification and reviews the farmer's notes
3. Consultation takes place (on-platform chat or scheduled call)
4. Expert submits written reply and marks consultation as completed
5. Farmer pays the total fee ($\text{hourly_rate} \times \text{duration_minutes} / 60$)
6. Admin receives commission from the total fee

2.6.3 Knowledge Contribution

- Experts can flag crop disease entries for updates
- May contribute to the crop encyclopedia with Admin approval

2.7 Tour Guide

Role Summary

Guides are local individuals hired by tourists during farm and agri-tourism visits. They are linked to Tour Bookings and provide language-specific guidance across farming areas. Guides earn a daily rate; admin takes a commission from the total booking.

2.7.1 Profile

- Register with languages spoken, current location, and daily rate
- Mark availability; rating built through completed tour bookings

2.7.2 Booking & Earnings

- Assigned to a TOUR_BOOKING when a tourist selects them during booking
- Guide fee paid as part of the booking payment; admin commission auto-deducted
- View upcoming assigned bookings in their dashboard

2.8 General User / Visitor

Role Summary

General Users are non-specialist members of the public — students, researchers, health-conscious individuals, or urban consumers. They can buy from the marketplace, explore the crop encyclopedia, use nutrition tools, and access cooking content without needing a specialised role.

- Browse and search the full Crop Encyclopedia
- Use Vitamin-based crop recommendation: choose a vitamin → see crop suggestions
- Use Season/Region/Soil-based crop suggestion tools
- View disease database with images, symptoms, and solutions
- Buy products from the marketplace (from Dealers)
- Use the Profit Calculator for research or planning purposes
- View country-wise crop demand and trade data
- Send consultation enquiries (if upgraded to farmer role)
- Access cooking recipes and nutrient retention guides

3. Working Principle — Module by Module

3.1 User Authentication & Role Management

Step	Process
Registration	User fills name, email, password, and selects a role (farmer/dealer/tourist/cook/expert/guide/user). Password stored as bcrypt hash.
Approval (Dealer/Farmer)	Admin reviews registration. Dealers require license verification. Farmers may require district validation before full access.
Login	Email + password → system validates hash → session created with role. Role determines which dashboard and menu items are shown.
Role-Based Routing	PHP session checks role on every page load. Unauthorised access redirected to login. Admin bypasses all role checks.
Password Reset	Email-based OTP or admin-initiated reset. Token stored temporarily in DB.

3.2 Crop Encyclopedia Module

The Crop Encyclopedia is the knowledge backbone of KrishiDisha. Every other module — disease detection, recommendations, marketplace, nutrition — is ultimately linked to this central CROP table.

3.2.1 Data Flow

- Admin adds a crop with: scientific name, origin, history, image URL, Bangladesh availability, trade status (export/import/both/none), and global demand index
- Admin links crop to: varieties (CROP_VARIETY), countries (CROP_COUNTRY via COUNTRY), vitamins (CROP_VITAMIN via VITAMIN)
- User opens the encyclopedia → sees crop cards with images
- User clicks a crop → full detail page loads via SQL JOIN across CROP, CROP_VARIETY, DISEASE, CROP_COUNTRY, CROP_VITAMIN
- User sees: scientific name, origin, history, BD availability, export/import status, varieties, vitamins, diseases, and related recipes

3.2.2 Search & Filter

- Search by crop name or scientific name (SQL LIKE query)
- Filter by: crop type (vegetable/fruit/grain/spice), season, Bangladesh availability, trade status
- Country filter: see which crops Bangladesh exports or imports with demand scores

3.3 Marketplace Module

How It Works — Summary

Farmer lists produce → Dealer browses and orders → Dealer stocks inventory → Dealer sells to User → Admin earns commission on Dealer sales. All payments tracked in a unified PAYMENT table.

3.3.1 Farmer → Dealer Flow

12. Farmer creates a PRODUCT listing: crop_id, quantity_kg, price_per_kg, description, image
13. Dealer browses products, filters by crop/region/price, places a MARKETPLACE_ORDER
14. Order status: pending → Farmer confirms → shipped → delivered
15. Payment made by Dealer to Farmer via selected method (bKash/Nagad/card/bank)
16. PAYMENT record created; DEALER_INVENTORY updated with new stock

3.3.2 Dealer → User Flow

17. Dealer sets sell_price (higher than buy_price) on their DEALER_INVENTORY entry
18. General User browses dealer listings and places a MARKETPLACE_ORDER
19. On order confirmation: total_amount calculated; admin_commission = total_amount × admin_rate
20. Dealer net = total_amount – admin_commission
21. ADMIN_COMMISSION record created, linked to PAYMENT and ADMIN

3.3.3 Commission Calculation

Example	Total Sale (BDT)	Admin Commission (5%)	Dealer Net
Tomato 100kg @ ₳40/kg	₳4,000	₳200	₳3,800
Rice 500kg @ ₳50/kg	₳25,000	₳1,250	₳23,750

3.4 Crop Disease Detection Module

22. Admin populates DISEASE table: disease_name, crop_id (FK), symptoms (text), solution (text), image_url, severity_level
23. Farmer or user opens Disease section and selects a crop
24. System retrieves all diseases for that crop via: `SELECT * FROM DISEASE WHERE crop_id = ?`
25. Displays: disease image, symptom list, recommended solution, and severity
26. Farmer can also search by symptom keyword (SQL LIKE across symptoms column)
27. Future: image upload → AI disease classification (planned extension)

3.5 Crop Recommendation System

3.5.1 Region/Soil/Season-Based

28. User selects: district (from REGION table), soil type, and current season
29. System queries `CROP_RECOMMENDATION JOIN CROP WHERE region_id = ? AND soil_type = ? AND season = ?`
30. Results ranked by suitability_score (highest first)
31. User sees recommended crops with images, scientific names, and cultivation tips

3.5.2 Vitamin/Nutrition-Based

32. User selects a vitamin (e.g., Vitamin C, Iron, Vitamin B12) from VITAMIN table
33. System queries `CROP_VITAMIN JOIN CROP WHERE vitamin_id = ? ORDER BY content_per_100g DESC`
34. User sees crops ranked by vitamin content, with amount per 100g shown
35. User can click a crop → see recipes that preserve that vitamin through optimal cooking method

3.6 Nutrition & Authentic Food Module

This module bridges the gap between agricultural produce and culinary health science. It serves both general users seeking healthy eating guidance and tourists seeking authentic Bangladeshi food experiences.

3.6.1 Recipe & Cook System

Actor	Action
Cook (registers)	Creates profile: speciality, location, price_per_session, marks available_for_tourists = true
Cook (adds recipe)	Creates RECIPE: crop_id, recipe_name, ingredients, cooking_steps, method, nutrient_retention_pct, is_authentic, image
Tourist (explores)	Browses recipes; filters by crop, cooking method, authentic flag, or vitamin retained
Tourist (orders)	Selects recipe + available cook → places TOURIST_FOOD_ORDER with special_request
Cook (fulfils)	Sees order in dashboard → confirms → prepares → marks as served
Admin (earns)	Commission % auto-deducted from amount_paid and logged in ADMIN_COMMISSION

3.6.2 Nutrient Retention by Cooking Method

Method	Avg. Retention	Best For
Raw / Uncooked	95–100%	Salads, fruits, vitamins C & B
Steaming	85–90%	Vegetables, retains minerals well
Boiling	60–75%	Grains, legumes; vitamins leach into water
Baking	70–80%	Root vegetables, complex carbohydrates
Frying	50–65%	Adds fat; destroys heat-sensitive vitamins

3.7 Agri-Tourism Module

The tourism module enables Farmers to open their lands as experiential destinations and Tourists to book guided farm visits. This module generates revenue for both farmers and guides while contributing commission to the Admin.

3.7.1 Tour Booking Workflow

36. Farmer registers FARM_LAND: name, location, area_acres, description, image, visit_fee, open_for_visit = true
37. Tourist browses available farm lands on an interactive listing page
38. Tourist selects a land, chooses visit_date, enters num_visitors
39. (Optional) Tourist browses GUIDE list: filter by language and rate; selects a guide
40. System creates TOUR_BOOKING: tourist_id, land_id, guide_id (nullable), visit_date, land_fee, guide_fee
41. Total = land_fee × num_visitors + guide_fee (daily); admin_commission auto-calculated
42. Payment made → booking status → confirmed
43. On visit date: farmer and guide receive confirmation. Status → visited after completion

3.7.2 Commission Flow — Tourism

Example Booking

Land visit fee: ₹500 × 3 visitors = ₹1,500

Guide daily rate: ₹800

Total booking: ₹2,300

Admin commission (8%): ₹184

Farmer receives: ₹1,316 (land fee minus admin %)

Guide receives: ₹800 (guide fee paid directly; separate admin cut may apply)

3.8 Expert Consultation Module

44. Farmer opens Consultation section → browses EXPERT list (filter by specialisation, rating, availability)
45. Farmer clicks 'Book Consultation' → selects expert, date, duration, and writes farmer_notes
46. CONSULTATION record created: status = pending
47. Expert logs in → sees pending consultations in dashboard
48. Expert reviews farmer notes, schedules or responds → writes expert_reply → marks status = completed
49. Total fee = hourly_rate × (duration_minutes / 60)
50. Farmer pays via chosen payment method → PAYMENT record created
51. Admin commission % deducted → ADMIN_COMMISSION record logged

3.9 Profit Calculator

52. User selects a crop from encyclopedia and enters: land area (acres), total investment (BDT), expected yield (kg), and current market price (BDT/kg)
53. System calculates: Gross Revenue = expected_yield_kg × market_price
54. Estimated Profit = Gross Revenue – investment
55. Result displayed with profit margin % and break-even analysis

56. Calculation saved to PROFIT_CALC table for history and future reference

57. User can compare profitability across multiple crops side by side

3.10 Unified Payment System

All financial transactions across every module (marketplace, consultation, tour booking, food orders) are recorded in the single PAYMENT table. This enables unified revenue reporting and commission tracking for the Admin.

Payment Method	Type	Use Case	Status Tracking
bKash	Mobile Banking	All modules	pending→success/failed
Nagad	Mobile Banking	All modules	pending→success/failed
Rocket	Mobile Banking	All modules	pending→success/failed
Credit/Debit Card	Card Payment	Marketplace, Tourism	pending→success/failed
Bank Transfer	Bank	Large dealer orders	pending→success/refunded

4. Database Design Summary

KrishiDisha uses a fully normalised relational database (MySQL) consisting of 26 tables across six functional domains. All tables use integer primary keys and foreign key constraints to enforce referential integrity.

Domain / Group	Tables
Users & Roles	USER, ADMIN, FARMER, DEALER, TOURIST, COOK, EXPERT, GUIDE
Crop Encyclopedia	CROP, CROP_VARIETY, DISEASE, CROP_COUNTRY, COUNTRY
Recommendations	REGION, CROP_RECOMMENDATION
Nutrition	VITAMIN, CROP_VITAMIN, RECIPE
Marketplace & Payments	PRODUCT, MARKETPLACE_ORDER, DEALER_INVENTORY, PAYMENT, ADMIN_COMMISSION
Tourism & Consultation	FARM_LAND, TOUR_BOOKING, CONSULTATION, TOURIST_FOOD_ORDER, PROFIT_CALC

4.1 Key Relationships

- USER is the parent entity for all roles (1:1 via user_id FK)
- CROP is the central entity linked to diseases, varieties, vitamins, recipes, products, recommendations, and profit calculations
- PAYMENT is a polymorphic table: reference_id + reference_type link to any payable entity
- ADMIN_COMMISSION references PAYMENT and ADMIN, enabling full revenue audit trail
- MARKETPLACE_ORDER links PRODUCT (from Farmer), BUYER (User), and DEALER in a single transaction record
- TOUR_BOOKING links TOURIST, FARM_LAND (Farmer's), and GUIDE in one booking record

4.2 Normalization

- 1NF: All tables have atomic values and a defined primary key
- 2NF: No partial dependencies; all non-key attributes depend on the full primary key
- 3NF: No transitive dependencies; country data separated into COUNTRY table; vitamin data in VITAMIN table
- Junction tables (CROP_VITAMIN, CROP_COUNTRY, CROP_RECOMMENDATION) resolve all many-to-many relationships

5. Technology Stack

Layer	Technology	Purpose
Database	MySQL 8.x	Relational data storage, joins, transactions, FK constraints
Backend	PHP 8.x	Session management, CRUD operations, business logic, commission calc
Frontend	HTML5 + CSS3	Page structure and styling
UI Framework	Bootstrap 5	Responsive layout, cards, modals, forms, tables
Local Server	XAMPP (Apache)	Local development environment: PHP + MySQL + Apache
Code Editor	Visual Studio Code	Development IDE with PHP/HTML/SQL extensions
Image Assets	Unsplash / Pexels	Free high-quality crop, farm, and food photography
Icons	Flaticon / Bootstrap Icons	UI icon set for navigation and feature indicators
Version Control	Git / GitHub	Source code management and collaboration

6. High-Level System Interaction Map

Platform Architecture Overview

All users interact through the Web Interface (PHP + HTML + Bootstrap).

The PHP backend handles all business logic, session control, and database queries.

MySQL stores all data across 26 normalised tables.

The Admin has a separate privileged dashboard for system governance.

Commission is automatically calculated and recorded on every financial transaction.

Actor	→	Action	→	System Response
Farmer	→	Lists produce	→	PRODUCT created; Dealer notified
Dealer	→	Orders from Farmer	→	MARKETPLACE_ORDER; inventory updated
Dealer	→	Sells to User	→	Order + admin commission auto-calculated
Tourist	→	Books farm tour	→	TOUR_BOOKING created; Guide notified
Tourist	→	Orders authentic food	→	TOURIST_FOOD_ORDER; Cook notified
Farmer	→	Requests consultation	→	CONSULTATION created; Expert notified
Expert	→	Completes consultation	→	Status updated; Farmer pays; Admin earns commission
Cook	→	Serves food order	→	Order marked served; Admin commission logged
Admin	→	Reviews dashboard	→	Full revenue, commission, and activity report
User	→	Uses profit calculator	→	PROFIT_CALC record saved; result displayed

7. Feasibility & Future Scope

7.1 Academic Feasibility

- All modules are implementable within a single academic semester using free tools (XAMPP, VS Code, MySQL, PHP, Bootstrap)
- The system covers all major DBMS concepts: normalisation (1NF–3NF), SQL joins, CRUD operations, transactions, foreign keys, many-to-many relationships via junction tables
- Role-based access control demonstrates real-world application of database-driven web systems
- Modular architecture allows incremental development: encyclopedia first, then marketplace, then tourism, etc.

7.2 Planned Future Enhancements

Feature	Description
AI Disease Detection	Image upload → convolutional neural network classifies crop disease automatically
Weather API Integration	Real-time weather data to enhance crop and season recommendations
Live Chat Consultation	WebSocket-based real-time chat between Farmer and Expert
Mobile App (Android/iOS)	React Native app for field use by farmers without desktop access
Map-Based Farm Browsing	Google Maps / Leaflet.js integration for tourist land discovery
AI Price Prediction	ML model to predict future crop prices based on historical market data
SMS Notifications	bKash/Nagad payment confirmation and booking alerts via SMS gateway
Multi-language Support	Bengali (বাংলা) interface alongside English for rural farmer accessibility

8. Conclusion

KrishiDisha represents a holistic approach to agricultural digitalisation in Bangladesh. By integrating eight distinct user roles, ten functional modules, twenty-six database entities, and a unified financial layer, the system addresses real-world pain points faced by farmers, traders, researchers, tourists, and nutritional learners.

The platform is uniquely positioned at the intersection of agriculture, commerce, education, tourism, and food science — all powered by a well-normalised relational database that ensures data integrity, efficient retrieval, and scalable growth.

Key Strengths of KrishiDisha

- ✓ Centralised crop intelligence with full scientific, economic, and nutritional data
- ✓ Fully functional marketplace with transparent commission tracking
- ✓ Agri-tourism ecosystem connecting farmers, guides, cooks, and tourists
- ✓ Role-based access control covering 8 distinct user types
- ✓ 26-table normalised database covering all DBMS course objectives
- ✓ Scalable architecture with clear pathways for AI and mobile extensions
- ✓ Built entirely with free, open-source tools suitable for academic deployment

This project is submitted as a complete DBMS course project demonstrating practical knowledge of relational database design, system architecture, multi-role web application development, and real-world problem solving in the agricultural domain of Bangladesh.

KrishiDisha — কৃষিদিশা

Empowering Bangladesh's Agriculture through Intelligent Data